01 VORTEX GRIP

Braided over/under helical straps for dynamic multi-point contact.

61.6 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



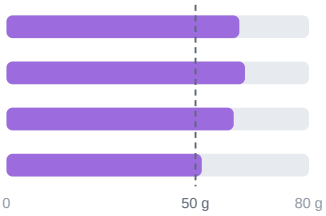
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.0 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.5 × 31.7 mm**
Peak texture height	**3.36 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 2.2 counterbore**
Material volume / surface	**49.67 cm³ / 437 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**61.6 g**
PETG	1.27 g/cm³	**63.1 g**
TPU	1.21 g/cm³	**60.1 g**
ABS	1.04 g/cm³	**51.7 g**



COLOURWAY



#3A225C #9C6CDE

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**22.6 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 20.7 m (100 % flow)**

DESIGN INTENT

Sixteen straps wound in opposite directions cross in an over/under braid, leaving dark valleys between them. Every crossing gives the hand an edge to bite on, so the grip works from any hand position on the handle.

Straps rise no faster than 45 degrees, so no supports are needed.

KEY FEATURES

- 8 right-hand + 8 left-hand straps on a 36 degree helix
- Over/under weave: the strap on top alternates at every crossing
- Valleys drop to about the 1.2 mm base wall, which keeps the mass low

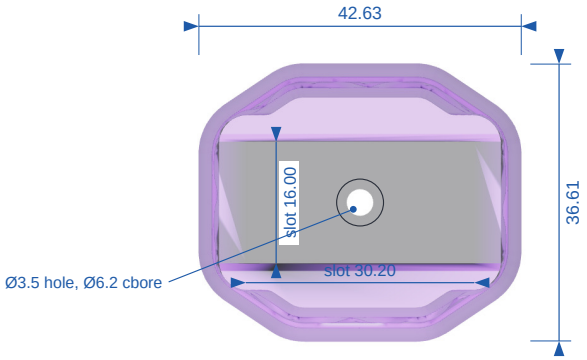
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.0 mm (30 + 0.2, 16 space)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 2.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.12 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.6 mm**

FILES

stl/01_vortex_grip.stl	19.4 MB · 387,216 triangles
3mf/01_vortex_grip.3mf	4.1 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

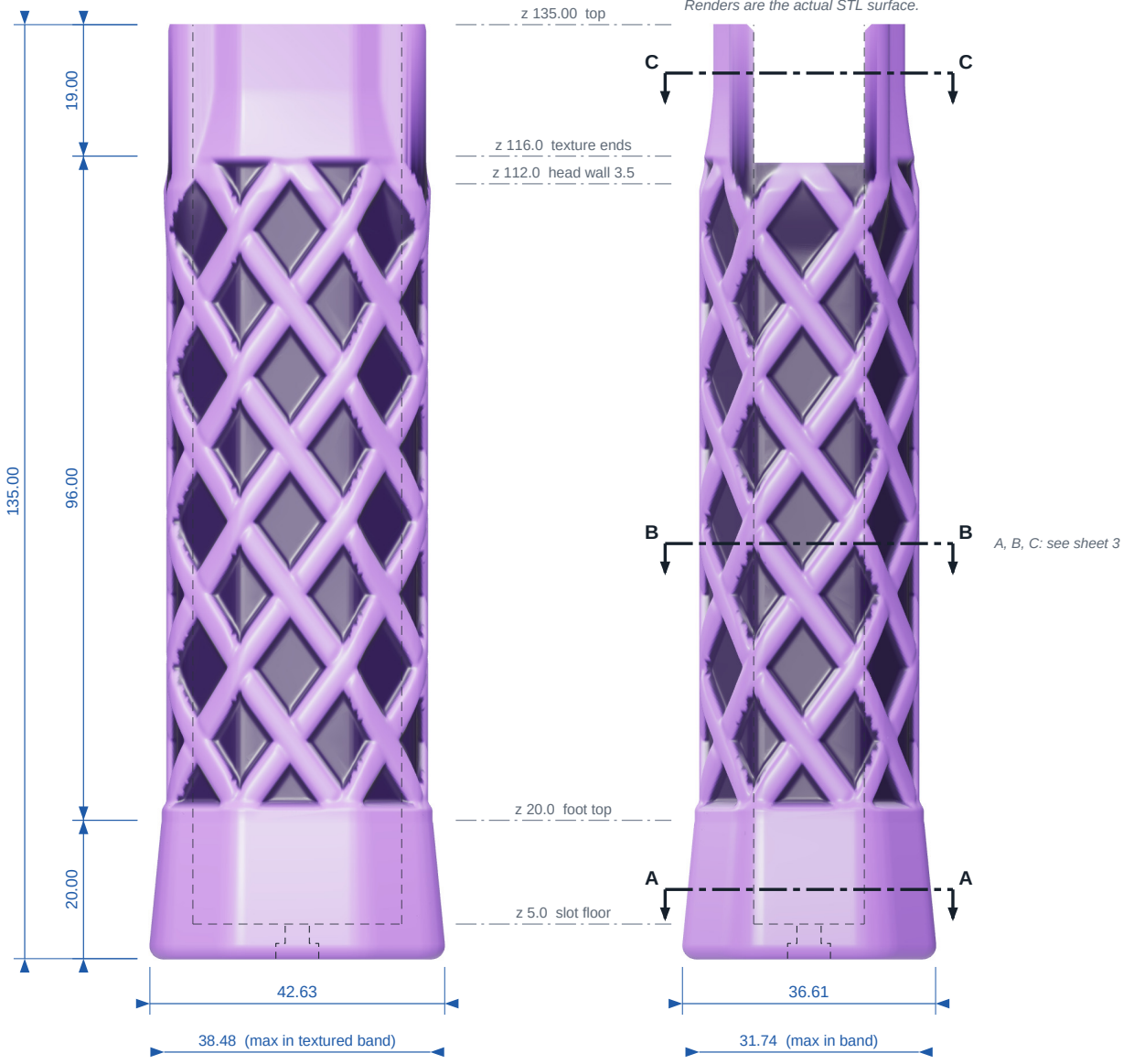


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.00**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.48 × 31.74**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.36**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 2.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

VORTEX GRIP

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

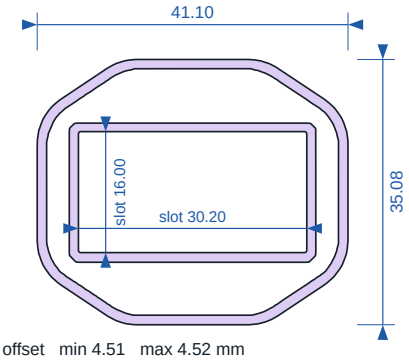
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 100 solid, end walls slit 20 mm.
4. Outer envelope matches PickleballGrip_1.3mf below z 90; head thickened.

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **61.6 g**	VOLUME **49.67 cm³**	SURFACE AREA **437 cm²**
REVISION **8**	DATE **2026-10-23**	SHEET **2 of 4**

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

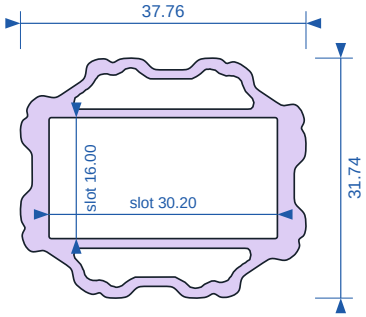
SECTION A–A z = 10 mm · foot



Skin offset min 4.51 max 4.52 mm

Area 277 mm² · outer perimeter 128.9 mm

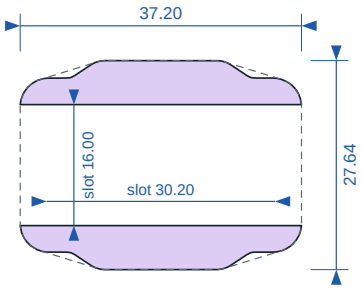
SECTION B–B z = 60 mm · textured band



Skin offset min 1.29 max 2.85 mm

Area 308 mm² · outer perimeter 118.0 mm

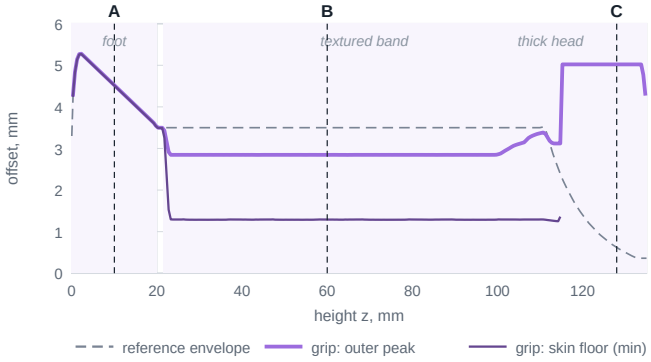
SECTION C–C z = 128 mm · rim



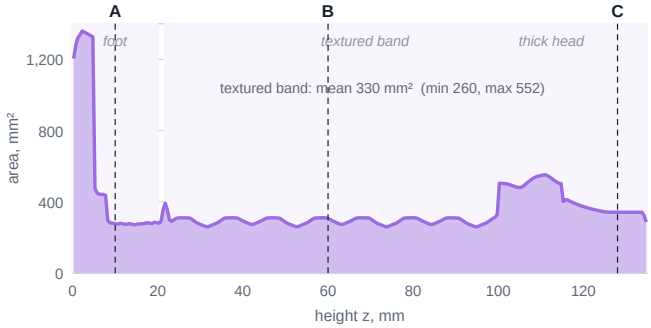
Skin offset max 5.02 mm

Area 342 mm² · outer perimeter 115.3 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.00 mm**
Slot area / perimeter	**483.1 mm² / 91.9 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 2.2**
Sealed cavity / min wall	**22.6 cm³ / 1.12 mm**
Outer W × D in band (max)	**38.48 × 31.74 mm**
Material volume / surface	**49.67 cm³ / 436.5 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck	116.0	112.0	**-4.0**
Head (solid, thick wall, slot opening)	112.0	135.00	**23.0**

Relief floor above reference line (design / min)	**1.20 / 1.29 mm**
Slot wall over the head (design / min)	**3.50 / 3.50 mm**
Peak texture height above base	**3.36 mm**

OUTER DIMENSIONS BY HEIGHT

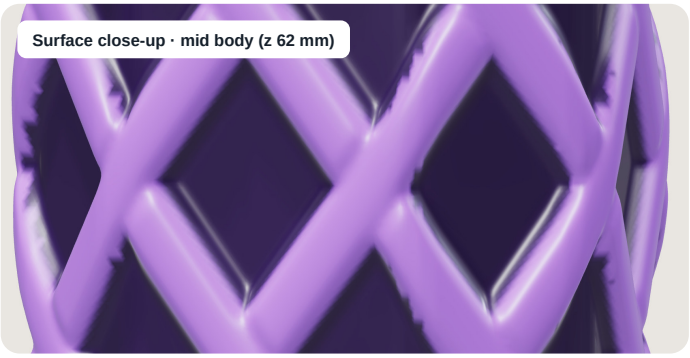
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	902	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	282	3.52	3.52
30.0	37.76	31.74	114.5	274	1.29	2.85
45.0	37.76	31.74	118.0	303	1.29	2.85
60.0	37.76	31.74	118.0	307	1.29	2.85
75.0	37.76	31.74	114.1	272	1.29	2.85
90.0	37.76	31.74	117.1	310	1.29	2.85
105.0	38.06	31.74	118.1	481	1.29	3.11
116.0	37.20	30.22	117.2	412	open	5.02
125.0	37.20	27.75	115.4	345	open	5.02
132.0	37.20	27.64	115.3	342	open	5.02

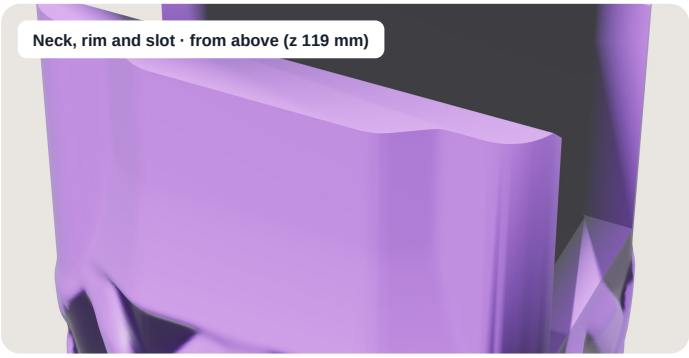
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

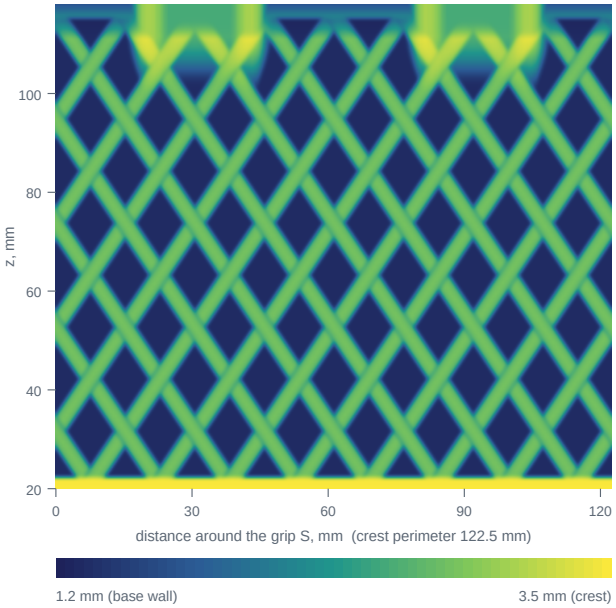


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Straps	**8 + 8, counter-wound**
Helix angle	**36 deg from vertical**
Strap pitch (around)	**15.31 mm**
Strap width	**6.1 mm around / 5.0 mm across**
Strap top height	**2.35 – 2.86 mm above base**
Valley floor	**1.29 mm above base**
Over/under step	**0.37 mm**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.12 mm**
Paddle slot	**30.2 × 16.0 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.6 mm**